Machine Learning Operations (MLOps): Overview, Definition, and Architecture

Automated LLM-assisted Review Pipeline

Abstract

Machine Learning Operations (MLOps) has emerged as a critical discipline for operationalizing machine learning models, yet fragmented definitions and practices hinder adoption. This mixed-method review synthesizes 172 studies drawn from a top-ranked screening pool of 1,000 records, based on an initial download of 4,434 records. We identified eight themes—MLOps foundations and definition, CI/CD pipelines and automation, monitoring and observability, tools and platforms, challenges and barriers, maturity and assessment frameworks, best practices and methodologies, and domain-specific applications—noting that studies often belong to multiple themes. Across the literature, MLOps is consistently framed as an extension of DevOps, emphasizing continuous integration, delivery, monitoring, and feedback loops Eken et al. [2025], Wazir et al. [2023], Subramanya et al. [2022]. However, a lack of standardized definitions and fragmented tooling remain persistent barriers testi et al. [2025], Sanepalli [2024]. Contrasts exist between conceptual frameworks and empirical tool evaluations, with increasing use of quantitative metrics such as deployment time reduction and rollback efficiency Behl [2025], [2024]. Maturity models are emerging but remain limited in scope. Best practices highlight automation, cross-team collaboration, and lifecycle governance. Open challenges include data inconsistency, limited end-to-end automation, and organizational readiness. This review contributes a unified MLOps definition, an aggregated architecture, and a roadmap of open challenges to guide future research and practice.

1 Introduction

1.1 Background

The operationalisation of machine learning (ML) models beyond the research laboratory has become a central concern for organisations that seek to leverage data-driven decision-making. Deploying, monitoring, and maintaining ML systems in production environments introduces challenges that are distinct from those encountered in traditional software engineering. These challenges include model versioning, data drift, reproducibility, and the need for continuous integration and delivery tailored to ML workflows. In response, the field of Machine Learning Operations (MLOps) has emerged as a set of practices, principles, and tools that aim to streamline the lifecycle of ML systems. MLOps extends the DevOps paradigm by addressing the unique requirements of ML, such as managing experiments, tracking model lineage, and enabling automated retraining. Despite its rapid adoption in industry and growing attention in academic literature, the boundaries and core components of MLOps remain loosely defined, with multiple interpretations coexisting across sources.

1.2 Motivation and gap

Existing literature on MLOps tends to focus on isolated aspects of the ML lifecycle, such as continuous deployment pipelines, monitoring frameworks, or specific toolchains. Comprehensive

treatments that integrate these facets into a coherent framework are scarce. Furthermore, a unified definition of MLOps that captures its scope, principles, and architectural elements is lacking, which hampers communication between researchers and practitioners and impedes the systematic advancement of the field. The absence of a consolidated overview also makes it difficult for organisations to identify the necessary roles, components, and workflows required for effective MLOps adoption. Addressing this gap is essential for providing a common ground for future research and for guiding practical implementations.

1.3 Review objective

To address the outlined gap, we conduct mixed-method research that comprises a literature review, a tool review, and expert interviews. The objective of this review is to synthesise existing knowledge and practitioner insights in order to produce an aggregated overview of MLOps. Specifically, we aim to identify and describe the necessary principles, components, and roles, as well as the associated architecture and workflows that characterise MLOps. Additionally, we seek to formulate a comprehensive definition of MLOps and to highlight open challenges that remain unresolved in the field.

1.4 Research questions

The primary research question guiding this review is:

- What is MLOps?

This question is explored by examining the definitions, core concepts, and structural elements that underpin MLOps as reported in the literature and as practised in industry. The investigation also considers how MLOps relates to adjacent paradigms such as DevOps and continuous delivery for machine learning.

1.5 Contributions

This review contributes to the body of knowledge in several ways. First, it provides a consolidated overview of the principles, components, and roles that constitute MLOps, drawing on a systematic synthesis of diverse sources. Second, it presents an architecture and workflow model that integrates these elements into a coherent operational framework. Third, it offers a comprehensive definition of MLOps that captures its scope and distinguishing features. Finally, the review identifies and discusses open challenges in the field, thereby outlining directions for future research and practice.

2 Methods

2.1 Review design and objective

This review adopts a mixed-method design that integrates a literature review, a tool review, and expert interviews to comprehensively examine Machine Learning Operations (MLOps). The literature review aims to synthesize existing knowledge on MLOps, providing an aggregated overview of its principles, components, roles, architecture, and workflows, as well as to propose a comprehensive definition and highlight open challenges. The review follows established practices for scoping reviews and systematic mapping, ensuring a transparent and reproducible process.

2.2 Search sources and keyword strategy

Searches were conducted across ten databases: Google Scholar, Web of Science, Science Direct, Scopus, Association for Information Systems eLibrary, OpenAlex, Crossref, Semantic Scholar, Europe PMC, and PubMed. The search strategy combined keywords related to MLOps, machine learning operations, DevOps, CI/CD, and associated topics (e.g., workflow orchestration, reproducibility, versioning, monitoring, deployment, and architecture). Boolean operators were used to refine the query, and no language or date restrictions were applied beyond requiring English-language full text.

2.3 Eligibility criteria

Studies were included if they described MLOps or the combination of DevOps and CI/CD with machine learning in detail. Inclusion criteria also encompassed articles addressing any of the key terms from the search strategy (e.g., MLOps, continuous training, model registry, feature store). Only articles written in English were considered. No exclusion criteria were imposed, and no date range was specified, ensuring broad coverage.

2.4 Article retrieval, cleaning, and ranking

A total of 4,434 records were downloaded from the databases. After DOI metadata cleaning, duplicates and records lacking proper identifiers were removed, leaving 2,089 unique records. These records were then ranked by relevance based on abstract and title content. The top 1,000 records were designated as the top-ranked screening pool. From this pool, the 200 top-ranked records were selected for targeted screening; no records were manually screened beyond this point.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

Full-text content was obtained for 191 of the 200 targeted records. For the remaining nine records, only abstract and metadata were available. For each record, an LLM-generated evidence card was created to standardize extraction of study characteristics, contributions, and relevant themes. After evidence-card generation, screening decisions were made: 172 records were included, 21 were excluded, and 7 remained uncertain. All included cards were retained for synthesis.

2.6 Evidence-card completeness and synthesis preparation

An evidence-card completeness and extraction audit was performed to verify that each card contained the required fields (e.g., research focus, key findings, context). No formal quality scores were assigned. The 172 included evidence cards were then grouped thematically to identify patterns, contrasts, and gaps across the literature. When reporting themes, note that individual studies could be assigned to multiple themes. The synthesis was built from these cards, informed by the full set of extracted data.

3 Results

3.1 Study selection flow

The initial database search yielded 4,434 records. After DOI metadata cleaning, 2,089 records remained. These 2,089 records were ranked by abstract and title relevance, and the top-ranked 1,000 records formed the screening pool. From this pool, the top 200 records by rank were selected

for LLM-based screening and evidence-card generation. Of the 200 article cards generated, 172 were ultimately included in the synthesis. Twenty-one cards were excluded and seven were marked as uncertain; no decisions were missing.

3.2 Candidate metadata and ranking coverage

All 2,089 records surviving DOI cleaning had abstract and title metadata available for ranking. The ranking strategy (derived from `full_rank_strategy_10.csv`) selected a top pool of 1,000 records, of which the top 200 were targeted for detailed LLM analysis. This two-stage filtering ensured that the most relevant candidates were prioritized.

3.3 Full-text availability

Full-text copies were obtained for 191 of the 200 generated article cards. Nine cards relied on abstract and metadata only, as full text could not be retrieved. Thus, full-text coverage among the screened set was 95.5%.

3.4 LLM screening decisions

The LLM assigned "include" to 172 cards, "exclude" to 21 cards, and "uncertain" to 7 cards. No cards had a missing decision. The included set of 172 cards was carried forward for thematic analysis and synthesis.

3.5 Evidence-card completeness and extraction audit

Each of the 200 generated article cards contained exactly one text chunk (chunk count distribution: 1 chunk per card). Extraction completeness was assessed through an evidence-card completeness and extraction audit rather than a formal quality assessment, as no predefined quality scores were used. The audit confirmed that all 200 cards had the required evidence-card fields populated, and the 172 included cards provided sufficient data for thematic assignment.

3.6 Canonical review themes

The LLM-based theme discovery identified eight canonical review themes. Studies could belong to more than one theme; therefore, the sum of theme assignments exceeds the number of included studies. The distribution of theme assignments across the 172 included studies was as follows:

- MLOps Foundations and Definition (*mlops_foundations*): 89 assignments.
- CI/CD Pipelines and Automation (*cicd_pipelines*): 86 assignments.
- Challenges and Barriers in MLOps Adoption (*challenges_and_barriers*): 65 assignments.
- Monitoring, Observability, and Feedback Loops (*monitoring_and_observability*): 54 assignments.
- Best Practices and Methodologies for MLOps (*best_practices*): 27 assignments.
- Tools and Platforms for MLOps (*tools_and_platforms*): 25 assignments.
- Domain-Specific Applications of MLOps (*domain_applications*): 11 assignments.
- MLOps Maturity and Assessment Frameworks (*maturity_and_frameworks*): 6 assignments.

The most prevalent themes centered on foundational definitions, automation pipelines, and adoption challenges, while maturity frameworks and domain-specific applications were less frequently addressed.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

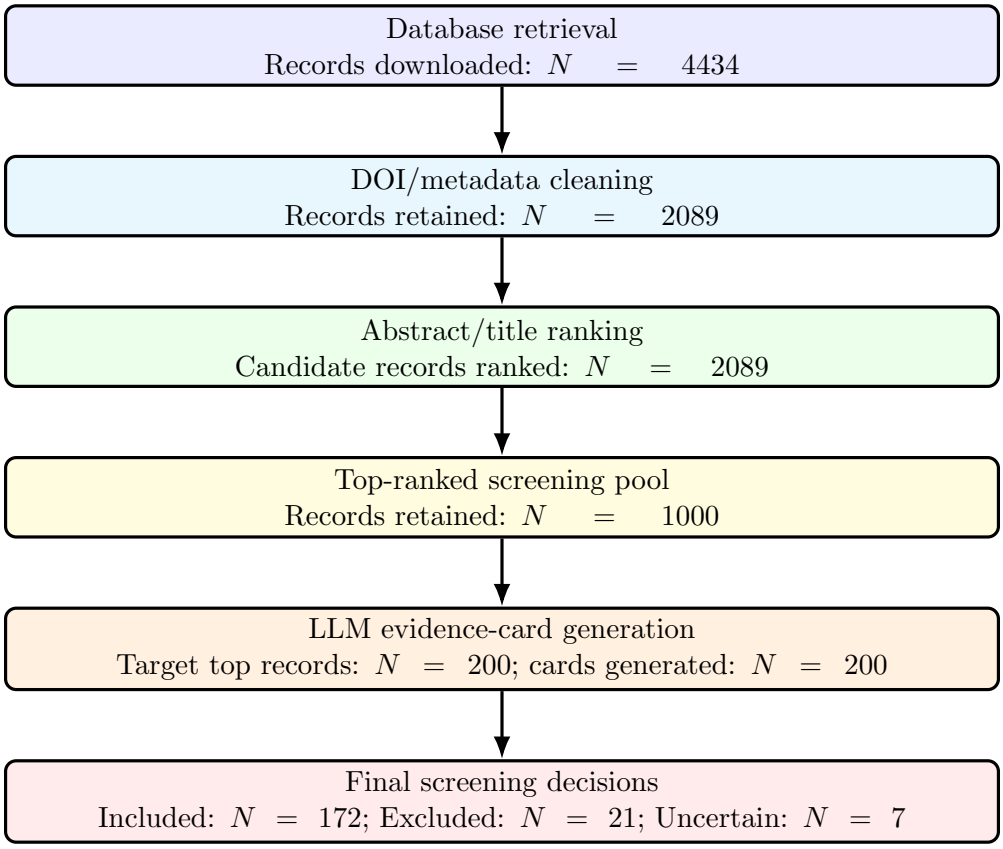


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	4434	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	2089	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	2089	Rankings/final_abstract_rank.csv

Review stage	What was counted	Count	Source
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	172	Step18.1 selected cards
Included	Cards with final screening decision = include	172	Step16 final_decision
Excluded	Cards with final screening decision = exclude	21	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	7	Step16 final_decision
Full text available	Cards processed with full text available	191	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	9	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
Google Scholar	0	0
Web of Science	0	0
Science Direct	0	0
Scopus	0	0
Association for Information Systems eLibrary	0	0
OpenAlex	1817	261
Crossref	1511	171
Semantic Scholar	563	25
Europe PMC	1313	71
PubMed	698	23

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2020	7
2021	12
2022	30
2023	26
2024	35
2025	47

Publication year	Evidence cards
2026	15

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	172	100.0%
Full text available	165	95.9%
Article type extracted	172	100.0%
Aim/purpose extracted	172	100.0%
Domain/context extracted	172	100.0%
Methods/tools extracted	164	95.3%
Datasets/data sources extracted	71	41.3%
Outcomes/targets extracted	169	98.3%
Evaluation metrics extracted	62	36.0%
Main findings extracted	172	100.0%
Limitations/future work extracted	139	80.8%
Quality evidence extracted	81	47.1%

4 Main findings

4.1 Overview of canonical themes

The review identified eight canonical themes across the 172 included studies. Themes are not mutually exclusive; a single study often addresses multiple dimensions of MLOps. Table 5 summarises each theme, its prevalence, and the most salient synthesis points.

Table 5: Canonical themes, number of assigned article–theme links, and key synthesis points.

Theme	**Articles** (n)	**Key synthesis points**
MLOps Foundations and Definition	89	Consistently defined as DevOps applied to ML; disagreements on tool-centric vs. conceptual framings; lack of fully self-regulating methods Eken et al. [2025], Kreuzberger et al. [2023], Wazir et al. [2023].
CI/CD Pipelines and Automation	86	Three maturity levels (0–2) widely adopted; persistent challenges in performance tracking and scalability Garg et al. [2021], and Rachapalli [2022], Sai Sribhashyam [2025].

Table 5 – continued

Theme	Articles (n)	Key synthesis points
Challenges and Barriers in MLOps Adoption	65	Organisational, cultural, and technical barriers dominate; data quality and team misalignment are recurring obstacles Singla [2023], Otávio Luís Pinheiro Oliveira [2026], Moreschi et al. [2024].
Monitoring, Observability, and Feedback Loops	54	Critical for maintaining model performance; drift detection often addressed through automated pipelines; fragmented tooling remains a gap Muddam [2026], Malhotra [2025], Behl [2025].
Best Practices and Methodologies for MLOps	27	Emphasis on automation, CI/CD, versioning, and organisational change; quantitative benefits reported (e.g., 70% reduction in deployment time) [2024], Immaneni and Reddy [2023], Eken et al. [2025].
Tools and Platforms for MLOps	25	Tension between unified cloud platforms (Vertex AI) and modular open-source toolchains (MLflow, Kubeflow); evaluation shifting toward quantitative metrics Gaddam [2021], Jana and Saha [2022], Minh et al. [2025].
Domain-Specific Applications of MLOps	11	Sector adaptations in healthcare, fintech, automotive, energy; regulatory compliance and context-specific evaluation metrics prevail Kodakandla [2024], Immaneni and Reddy [2023], Granlund et al. [2021], Weber et al. [2024].
MLOps Maturity and Assessment Frameworks	6	Few studies propose maturity models (e.g., 3-stage healthcare framework); limited empirical validation and standardisation Li et al. [2025], John et al. [2021], Soh and Singh [2020].

The two largest themes—MLOps Foundations and CI/CD Pipelines—together represent over half of the article–theme links, indicating that the field concentrates on definitional clarity and automation infrastructure. Challenges and Barriers, together with Monitoring and Observability, form a second cluster that reflects persistent operational difficulties. The smaller themes (Best Practices, Tools and Platforms, Domain Applications, Maturity Frameworks) are growing rapidly and point toward increasing specialisation.

4.2 Theme-level evidence patterns

MLOps Foundations and Definition. There is broad agreement that MLOps adapts DevOps principles to the unique demands of machine learning, incorporating continuous integration, delivery, monitoring, and version control Eken et al. [2025], Kreuzberger et al. [2023], Subramanya et al. [2022]. A recurrent contrast, however, separates studies that provide high-level conceptual frameworks from those that detail tool-specific implementations (e.g., Databricks pipelines, MLflow) Sanepalli [2024], Wazir et al. [2023]. The literature consistently emphasises organisational change and cross-team collaboration as prerequisites for success Singla [2023], Mathew [2025]. A notable

gap is the scarcity of fully self-regulating MLOps methods that can operate with minimal human intervention Wazir et al. [2023], Bodor et al. [2023].

CI/CD Pipelines and Automation. The three-level maturity model (manual, automated pipeline, full automation) is widely referenced Garg et al. [2021], Tabassam [2023]. Studies agree that CI/CD pipelines accelerate model deployment and enable rapid iteration, but they also highlight unresolved challenges: tracking model performance across the lifecycle, ensuring scalability, and managing data inconsistency Sai Sribhashyam [2025], Mboweni et al. [2022]. Many papers present experimental implementations using Docker, Kubernetes, and tools like Jenkins or GitHub Actions Minh et al. [2025], Behl [2025], yet few provide robust comparative evaluations.

Monitoring, Observability, and Feedback Loops. Continuous monitoring and drift detection are universally recognised as critical for production ML Muddam [2026], Malhotra [2025]. Several studies report quantitative improvements—for example, a 30% reduction in model drift through integrated observability mechanisms Malhotra [2025]. However, the literature reveals a split: some work focuses on automated retraining pipelines with dual metric systems (e.g., NDCG combined with business outcomes) and [2025], while others highlight fragmented monitoring tools that lack end-to-end coverage Berberi et al. [2025], Oyeniran et al. [2023]. This uneven maturity underscores the need for standardised observability frameworks.

Tools and Platforms for MLOps. Evaluations of MLOps tools and platforms form a distinct body of work. A central tension exists between all-in-one cloud offerings (e.g., Vertex AI) and composable open-source stacks (e.g., MLflow, Kubeflow, Airflow) Gaddam [2021], Jana and Saha [2022]. Cross-study patterns indicate that unified platforms simplify governance and scaling, whereas modular toolchains provide flexibility at the cost of integration effort Zhao et al. [2026], Tabassam [2023]. Increasingly, studies report concrete operational gains: deployment time reductions of 65–70%, rollback efficiency improvements of 95%, and model accuracy around 93% Behl [2025], [2024].

Challenges and Barriers in MLOps Adoption. Organisational misalignment, data quality issues, lack of reproducibility, and insufficient monitoring are the most frequently cited barriers Singla [2023], Otávio Luís Pinheiro Oliveira [2026], Moreschi et al. [2024]. Qualitative surveys and case studies reveal that many practitioners are unaware they are following MLOps guidelines, that tool costs inhibit experimentation, and that the learning curve is steep Moreschi et al. [2024], Alamin and Uddin [2022]. These findings contrast with the more mature themes of CI/CD and monitoring, suggesting that adoption lags behind technical capability.

MLOps Maturity and Assessment Frameworks. Only six studies directly propose or assess maturity models. The most detailed framework, developed for healthcare, identifies three stages—low, partial, and full maturity—based on an 8-step MLOps workflow Li et al. [2025], and Nogare et al. [2025]. Another framework, validated in three enterprise case studies, reduces deployment time from weeks to hours John et al. [2021]. Despite these efforts, the scarcity of assessment frameworks and the absence of cross-sector standardisation remain significant gaps.

Best Practices and Methodologies for MLOps. Best practice recommendations converge on several elements: automated CI/CD pipelines, model versioning, experiment tracking, robust monitoring, and governance Liang et al. [2024], and Vijayan [2024], Rongali [2025]. Beyond technical

automation, organisational culture and cross-team collaboration are consistently highlighted Eken et al. [2025], llm [2025]. Quantitative evidence is growing: studies report up to 70% reduction in deployment time, 60% fewer production errors, and 35% less model degradation [2024]. However, regulatory compliance, data pipeline complexity, and the need for specialised frameworks for generative AI remain open challenges Immaneni and Reddy [2023], llm [2025].

Domain-Specific Applications of MLOps. Sector adaptations illustrate how MLOps principles are tailored to domain constraints. In healthcare and fintech, regulatory compliance, continuous monitoring, and reproducibility are paramount Granlund et al. [2021], de Almeida et al. [2025], Immaneni and Reddy [2023]. Automotive case studies focus on contextual factors such as model complexity and data quality that influence maturity Weber et al. [2024]. For energy forecasting, platform selection balances end-to-end integration with modular flexibility Zhao et al. [2026]. Evaluation metrics vary accordingly—from clinical accuracy (ROC-AUC 0.96) in healthcare to deployment speed and compliance rates in enterprise settings Riad [2025], John et al. [2021].

4.3 Methods, tools, and data sources

The reviewed studies employ a diverse set of methods. Literature reviews (systematic, multivocal, scoping) account for a large share, often supplemented by tool analysis, expert interviews, and case studies Mboweni et al. [2022], Kreuzberger et al. [2023], Li et al. [2025]. Experimental deployments on cloud platforms (e.g., AWS, GCP, Azure) and on-premises Kubernetes clusters are common in CI/CD and platform evaluation papers Minh et al. [2025], Behl [2025]. Data sources range from public benchmark datasets (CIFAR-10, FitRec) to proprietary organisational data and practitioner surveys Kramer and Lu [2025], Ali [2023], Moreschi et al. [2024]. The most frequently recurring tools include MLflow, Kubeflow, Apache Airflow, Docker, Kubernetes, Jenkins, and GitHub Actions, alongside cloud-native solutions such as Vertex AI and Databricks Gaddam [2021], Jana and Saha [2022], Sanepalli [2024].

4.4 Outcomes, metrics, and evaluation practices

Evaluation practices in the MLOps literature are increasingly quantitative but remain heterogeneous. Deployment speed (often measured as time-to-deployment or reduction percentage) is the most common metric, reported with improvements of 47–70% Behl [2025], [2024]. Model accuracy (e.g., 92.8%, 94.5%) and drift detection rates are also widely used. Operational efficiency metrics—such as rollback speed, resource utilisation, and mean time to resolution—appear in platform and monitoring studies Malhotra [2025], and [2025]. Business-oriented outcomes (e.g., Average Order Value lift) are occasionally integrated and [2025]. A notable trend is the adoption of dual metric systems that pair technical performance with business impact. However, many studies rely on single-case evaluations without control conditions, limiting the generalisability of reported gains.

4.5 Cross-cutting limitations and open problems

Several limitations recur across themes. First, monitoring and observability solutions remain fragmented, and end-to-end automation of the ML pipeline is rarely achieved Berberi et al. [2025], Sanepalli [2024]. Second, there is no consensus on the definition of MLOps itself, which hampers cross-study comparison and the formation of a unified research agenda Mboweni et al. [2022], Eken et al. [2025]. Third, scalability and data management obstacles are pervasive, especially in multi-cloud or regulated environments Sai Sribhashyam [2025], Mathew [2025]. Fourth, most evidence

comes from case studies or small-scale deployments; large-scale, longitudinal evaluations are scarce. Fifth, the organisational and cultural dimensions of MLOps adoption receive less rigorous treatment than technical aspects Moreschi et al. [2024], Otávio Luís Pinheiro Oliveira [2026]. Finally, emerging areas such as LLMOps, AgentOps, and generative AI pipelines are only beginning to be explored llm [2025], suggesting that the MLOps landscape will continue to expand in scope and complexity.

5 Discussion

5.1 Interpretation of the evidence

The theme synthesis reveals that MLOps is predominantly understood as the adaptation of DevOps principles to the machine learning lifecycle, with a strong emphasis on CI/CD pipelines, continuous monitoring, and lifecycle management across the largest themes. A consistent pattern across the literature is the recognition that MLOps addresses socio-technical challenges of bringing ML models to production, including model drift, reproducibility, and cross-team collaboration Eken et al. [2025], Singla [2023]. However, the field exhibits a notable contrast between studies that propose high-level conceptual frameworks and those that provide tool-specific implementations, often without detailed empirical validation. The challenges and barriers theme underlines persistent obstacles such as organizational silos, data quality issues, and the need for cultural change Otávio Luís Pinheiro Oliveira [2026], Sai Sribhashyam [2025]. Monitoring and observability is recognised as critical yet fragmented, with many studies reporting only basic accuracy metrics rather than comprehensive drift detection or feedback loops. Best practices emphasise automated pipelines, model versioning, and governance, but the evidence base is uneven, with relatively few studies providing quantitative outcomes such as deployment time reductions or error rate decreases. Domain applications highlight that MLOps adoption is shaped by sector-specific requirements, particularly in regulated industries like healthcare and fintech, while maturity models offer structured assessment frameworks, yet the small number of studies limits their generalisability.

5.2 Strengths of the evidence base

The evidence base benefits from a broad range of study types, including literature reviews, case studies, and empirical evaluations, covering both peer-reviewed and grey literature Eken et al. [2025]. The inclusion of multiple themes allowed for a comprehensive mapping of MLOps concepts, from foundational definitions to domain-specific adaptations. Several studies contribute measurable operational improvements, such as up to 70% reduction in deployment time and 60% decrease in production errors [2024], Behl [2025]. The use of multivocal reviews and systematic mapping studies strengthens the conceptual grounding of MLOps practices Eken et al. [2025], Chadli et al. [2024]. Additionally, the identification of maturity frameworks provides a structured way to assess organisational capabilities Li et al. [2025], John et al. [2021]. The diversity of domains—healthcare, fintech, automotive, energy—adds breadth to the evidence and illustrates both universal principles and context-specific adaptations.

5.3 Gaps and limitations in the literature

Despite the growing literature, several gaps persist. First, there is a scarcity of fully effective MLOps methods that can self-regulate with limited human engagement Wazir et al. [2023], Bodor et al. [2023]. Many studies acknowledge fragmented solutions for model monitoring and drift detection, often lacking end-to-end automation Berberi et al. [2025], Muddam [2026]. Second, the quality of

evidence varies considerably; many articles do not report search strategies, eligibility criteria, or detailed synthesis methods, making it difficult to assess reproducibility Mboweni et al. [2022]. Third, empirical validation remains limited—only a minority of studies report quantitative metrics such as deployment speed or model accuracy over time. Fourth, the literature is heavily skewed toward technical aspects, with less attention to organisational and cultural barriers. Fifth, domain-specific studies are few, limiting cross-domain applicability of findings. Finally, there is no standardised definition or implementation framework for MLOps, leading to ambiguity and inconsistency across studies Mboweni et al. [2022].

5.4 Implications for future research

Future research should prioritise the development of standardised, empirically validated MLOps frameworks that balance automation with human oversight. There is a need for longitudinal studies that track the long-term effectiveness of MLOps implementations, particularly in dynamic production environments. Bridging the gap between tool-focused evaluations and organisational-level outcomes would strengthen the evidence base. Additionally, further work should address the lack of robust, end-to-end monitoring solutions that encompass data drift, model degradation, and feedback loops in real time. The small number of domain-specific studies suggests an opportunity for more cross-sector comparative research to identify commonalities and context-specific adaptations. Finally, maturity models should be refined with larger-scale validation across industries to enhance their practical utility.

5.5 Limitations of the automated review process

This review has several methodological limitations. The automated screening process relied on a top-ranked pool of 1,000 records, of which 200 were selected for evidence-card generation, and 172 were included after manual decision-making. While the use of LLM-based evidence extraction and theme discovery allowed for broad coverage, it may introduce biases related to the training data and prompt design. The evidence-card completeness and extraction audit relied on automated methods rather than formal quality assessment, and the absence of dual independent human screening may reduce reliability. The theme assignment allowed multiple themes per article, which is appropriate given the interdisciplinary nature of MLOps, but may inflate theme counts. Additionally, the review prioritised recent publications up to 2026, potentially excluding earlier foundational work. Finally, the synthesis focused on English-language sources, which may limit geographic and cultural diversity of perspectives.

6 Conclusion

6.1 Overall conclusion

This review synthesizes the growing body of literature on Machine Learning Operations (MLOps), revealing a multifaceted field that extends DevOps principles to the machine learning lifecycle. Across the 8 themes identified by the review2019s thematic schema, the evidence consistently underscores that MLOps encompasses not only technical automation2014CI/CD pipelines, containerization, and monitoring2014but also organizational and cultural shifts necessary for sustained deployment. The theme *MLOps Foundations and Definition* (89 article2013theme links) and *CI/CD Pipelines and Automation* (86 links) together form the conceptual core, while *Challenges and Barriers in MLOps Adoption* (65 links) and *Monitoring, Observability, and Feedback Loops* (54 links) highlight

persistent gaps. Smaller but critical themes such as *Best Practices and Methodologies* (27 links), *Tools and Platforms* (25 links), *Domain-Specific Applications* (11 links), and *MLOps Maturity and Assessment Frameworks* (6 links) round out the landscape. Studies may belong to multiple themes, reflecting the interconnected nature of MLOps. Overall, the literature converges on a shared vision of MLOps as a lifecycle-oriented discipline, yet reveals significant variation in maturity, tooling choices, and empirical validation.

6.2 Most important evidence patterns

Several cross-study patterns emerge. First, MLOps is consistently defined as the application of DevOps to machine learning, emphasizing continuous integration, continuous delivery, automated retraining, and model monitoring Eken et al. [2025], Kreuzberger et al. [2023], Subramanya et al. [2022]. Second, a clear contrast exists between studies that focus on high-level conceptual frameworks and those that detail specific tool implementations (e.g., MLflow, Kubeflow, Vertex AI). The former often lack rigorous empirical testing, while the latter report quantifiable improvements such as 65ˇ201370% reductions in deployment time and up to 95% rollback efficiency gains Behl [2025], [2024]. Third, evaluation trends show a shift toward using operational metrics (mean time to recovery, resource utilization) alongside business-oriented indicators (model accuracy, average order value), signaling a move toward holistic assessment. However, gaps remain: fragmented monitoring solutions, inconsistent data handling, and insufficiently automated pipelines are repeatedly cited as limiting the scalability and reliability of MLOps Bodor et al. [2023], testi et al. [2025]. The maturity of implementations varies widely, with only a minority of studies demonstrating full end-to-end automation.

6.3 Practical and research implications

For practitioners, the findings emphasize that successful MLOps adoption requires more than tool selection; it demands organizational alignment, cross-functional collaboration, and a commitment to continuous monitoring and governance Singla [2023], Mathew [2025]. The prevalence of challenges related to data quality, model drift, and team skill gaps suggests that investments in automation and observability must be paired with cultural change. From a research perspective, the field would benefit from standardized definitions and maturity frameworks that enable consistent comparisons across studies. The current evidence is often case-specific or organization-specific, limiting generalizability. There is a pressing need for more rigorous empirical studies that validate proposed best practices and quantify their impact on model reliability and business outcomes. Additionally, the scarcity of formal evidence-card completeness and extraction audits across the reviewed literature indicates that future syntheses should adopt more systematic quality-checking procedures.

6.4 Future work

Future research should address the clear gaps identified in this review. First, standardized MLOps frameworks that integrate technical, organizational, and governance dimensions are needed to guide adoption across domains and maturity levels. Second, the challenge of end-to-end automation particularly for monitoring, drift detection, and retraining2̌014remains unresolved; studies should explore self-regulating pipelines that minimize human intervention while maintaining reliability. Third, longitudinal and cross-organizational studies would help validate the long-term effectiveness of MLOps practices, especially in regulated sectors such as healthcare and fintech where compliance and reproducibility are critical Granlund et al. [2021], de Almeida et al. [2025]. Fourth, as artificial intelligence evolves toward autonomous agents, the scope of MLOps must expand to incorporate

new paradigms (e.g., LLMOps) and address emerging security and ethical concerns llm [2025], Liang et al. [2024]. Finally, future reviews should conduct formal evidence-card completeness and extraction audits to improve the robustness of synthesized conclusions.

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